Healthcare Data Visualization

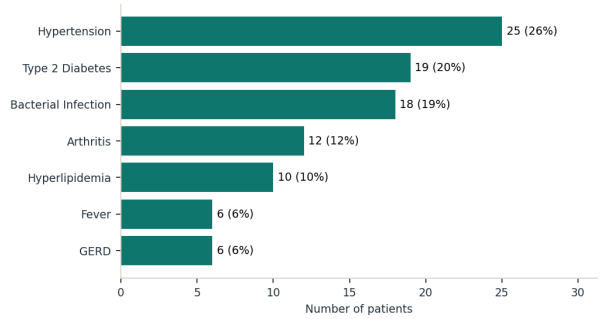
Disease, age, gender and medication patterns from the cleaned dataset

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Purpose. To turn the cleaned dataset (96 patients) into charts so that patterns in diseases, age, gender and medicines are easy to see. Charts were made in Python (pandas and matplotlib). The data is fictional practice data.

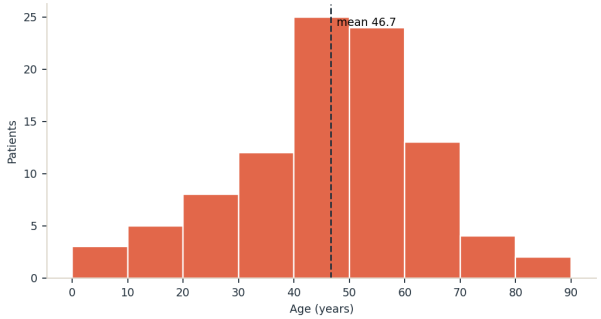
96	46.7	Hypertension	Metformin	49M / 43F
Patients	Average age	Top disease	Top medication	Gender count

1. DISEASE DISTRIBUTION



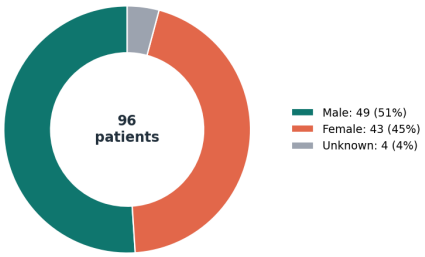
Hypertension is the largest group (25 patients), then Type 2 Diabetes (19) and Bacterial Infection (18). Chart type: horizontal bar, best for comparing categories.

2. AGE DISTRIBUTION



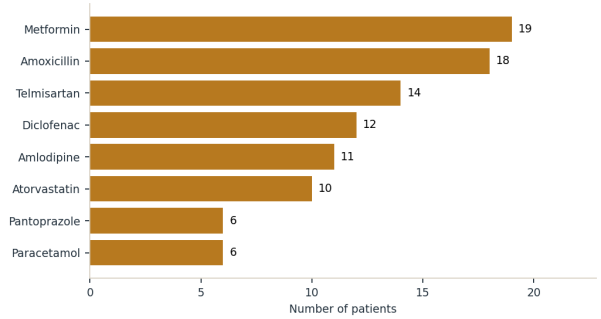
Most patients are middle-aged. Average age is 46.7 and the median is 49. Chart type: histogram, shows how ages are spread.

3. GENDER RATIO



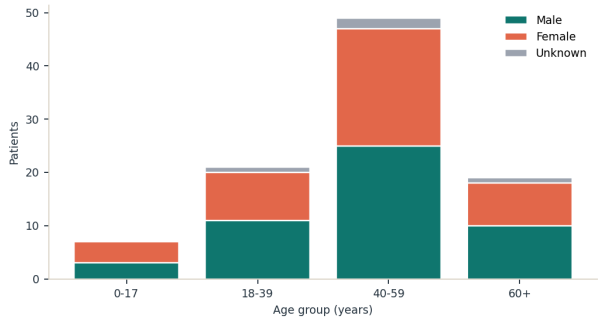
Male 49, Female 43, Unknown 4. The split is nearly even. Chart type: donut, shows share of a whole.

4. MEDICATION USAGE



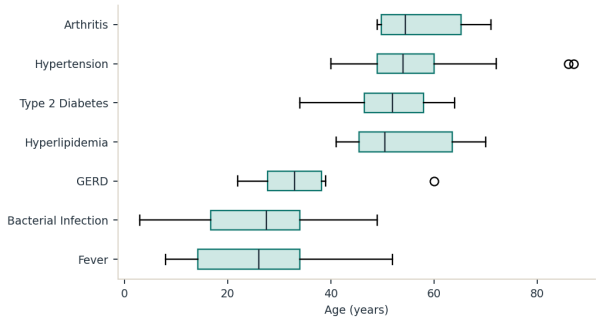
Metformin is used most (19 patients), matching diabetes cases. Chart type: horizontal bar.

5. AGE GROUPS BY GENDER



The 40-59 group is the largest (49 of 96 patients). Males and females follow the same pattern. Chart type: stacked bar.

6. AGE BY DISEASE



Chronic diseases affect older patients; infection and fever affect younger ones. Chart type: box plot (middle line = median).

KEY INSIGHTS

- **Disease burden:** Chronic conditions (hypertension, diabetes, arthritis, hyperlipidemia) account for 66 of 96 patients.
- **Age pattern:** Chronic-disease patients average 54.9 years; patients with infection or fever average 26.8 years.
- **Gender:** The ratio is close to even, so gender is not a major factor in this sample.
- **Medication link:** Medicine usage follows disease counts (for example, Metformin 19 for Type 2 Diabetes 19), as expected for first-line treatment.
- **Use in healthcare:** Such charts help plan clinic resources, spot high-risk age groups and monitor medicine use. Note: the sample is small and fictional, so the trends are for practice, not conclusions about real patients.

Source file: healthcare_dataset_cleaned.csv (output of the data cleaning task). Fictional data for internship practice only.